

A RESIDUE-DEPENDENT SOLVABILITY OBSTRUCTION FOR SKEW-BRACE EXTENSIONS:

$\mathrm{im} \varphi^+ = \ker o_\nu$, AND THE COUPLING SPLITS

MACBETH

ABSTRACT. Fix a skew brace H , an abelian group $(D, +)$ carrying a (possibly non-trivial) brace structure, and a good triplet $(\nu, \mu, \widehat{\sigma})$ of automorphism data. The skew-brace extensions $0 \rightarrow D \rightarrow G \rightarrow H \rightarrow 0$ inducing the triplet are classified by a second cohomology $H_{\mathrm{sb}}^2(H, D)$ whose classes are pairs $([\beta], [\bar{\tau}])$ bound by the Rathee–Yadav coupling. We ask which additive classes $[\beta] \in H_+^2$ arise, i.e. we compute the image of the additive forgetful map φ^+ . We show this image is the kernel of a single *residue-dependent* obstruction o_ν : when D is a trivial brace, o_ν is a group homomorphism on H_+^2 and $\mathrm{im} \varphi^+ = \ker o_\nu$. For a non-trivial kernel the governing equation splits into a *kernel coupling* and a *section coupling*; the kernel coupling is affine in β , with offset the additive coboundary of the intrinsic defect $\delta_h = \nu_h \widehat{\sigma}_h - \mathrm{id}$, and $\mathrm{im} \varphi^+$ is a torsor over $\ker o_\nu$ that misses the split class exactly when some defect section fails to be a homomorphism. These results correct an earlier, false assertion that the residue $[\nu]$ and the decomposition class are independent coordinates; the correction is the paper’s organising thread. The homomorphism law, the kernel coupling, and the torsor dichotomy hold for general H ; the biconditional for a non-trivial kernel is proved for $H = \mathbb{Z}/2$ and verified computationally for all 34 triplets with $H \in \{\mathbb{Z}/2, \mathbb{Z}/3\}$. Two core finite facts—a $|G| = 16$ lock and the non-injectivity of the forgetful map—carry axiom-free Lean 4 certificates.

1. INTRODUCTION

The problem. A *skew left brace* is a set B with two group operations $(B, +)$ and $(B, \circ)$ sharing a unit and satisfying $a \circ (b + c) = (a \circ b) - a + (a \circ c)$. Skew braces are the algebraic home of set-theoretic solutions to the Yang–Baxter equation and, through the Hopf–Galois correspondence, of Galois module structure; their extension theory is correspondingly central. Given a brace ideal $D \trianglelefteq G$ with quotient $H = G/D$, one wants to classify all extensions $0 \rightarrow D \rightarrow G \rightarrow H \rightarrow 0$ realising a fixed action of H on D . Rathee and Yadav [1] solved this: such extensions are classified by a second cohomology group $H_{\mathrm{sb}}^2(H, D)$, whose cocycles are *pairs* $(\beta, \bar{\tau})$ —an additive factor set β and a multiplicative factor set $\bar{\tau}$ —subject to a coupling equation, their (3.10).

The coupling is what makes skew-brace cohomology genuinely richer than group cohomology: the two factor sets cannot be prescribed independently. This paper studies one consequence. The *additive forgetful map*

$$\varphi^+ : H_{\mathrm{sb}}^2(H, D) \longrightarrow H_+^2 := H_{\mathrm{Gp}}^2((H, +), (D, +); \mu), \quad [(\beta, \bar{\tau})] \mapsto [\beta],$$

records only the additive part of an extension—concretely, the isomorphism type of the underlying additive group $(G, +)$. Which additive classes occur? That is, what is $\mathrm{im} \varphi^+$?

A false start, and why it is instructive. This question has an attractive wrong answer. The residue $[\nu]$ (the restriction of the brace action to the ideal) and the additive class $[\beta]$ look like independent coordinates: one governs the action, the other the additive extension. An earlier draft of this programme asserted exactly that—a product decomposition $[\nu] \perp [\beta]$ —and deduced

that every additive class is realisable for every residue. The argument reduced solvability of the coupling to a homogeneous, residue-free linear system by “reading $\circ$ off from $+$ and λ .”

The assertion is false, and the reduction is circular: reading $\circ$ off from $+$ and λ presupposes a brace action λ with the prescribed residue, which is precisely the existence question. A referee report of Rick’s [3], confirmed by an independent from-scratch reproduction, exhibited the coupling explicitly at $|G| = 8$ and $|G| = 16$: the residue *does* constrain which additive classes occur. We state this refutation plainly in §3; it is not an embarrassment to be buried but the hinge of the paper, because the correct object falls out of understanding exactly why the product fails.

The result. The correct object is a single obstruction class. Write $B_+^2 \leq Z_+^2$ for the additive coboundaries and $Z_\circ^2$ for the multiplicative cocycles (§2).

Theorem 1.1 (Trivial kernel; §4). *Let D be a trivial brace and fix a good triplet with ν a homomorphism. There is a $\mathbb{Z}$ -linear operator Q and a subgroup $A(Z_\circ^2) \leq C^3$, both built from the triplet alone, such that*

$$o_\nu: H_+^2 \longrightarrow C^3/A(Z_\circ^2), \quad [\beta] \mapsto [Q\beta],$$

is a well-defined group homomorphism, and $\text{im } \varphi^+ = \ker o_\nu$. Equivalently, an additive class $[\beta]$ extends to a full skew-brace cocycle inducing the triplet if and only if $o_\nu([\beta]) = 0$.

For a non-trivial kernel the obstruction acquires an affine offset, and the governing equation splits cleanly in two.

Theorem 1.2 (Non-trivial kernel; §5). *Let $(D, +)$ carry a brace structure λ^D with $(D, \circ)$ abelian and $(G, +)$ abelian, and let $\delta_h := \nu_h \widehat{\sigma}_h - \text{id}$ be the intrinsic defect. The skew-brace axioms for an assembly on $D \times H$ decompose as a kernel coupling*

$$(\star) \quad (\lambda_d^D - \text{id}) \beta(k, k') = \delta_k(d) + \delta_{k'}(d) - \delta_{k+Hk'}(d) \quad (d \in D, k, k' \in H)$$

and a section coupling (F), the latter being the non-trivial-kernel form of the Rathee–Yadav coupling. The kernel coupling is affine in β with β -independent offset the additive coboundary $c^d = d_+[\delta_\bullet(d)]$; and the split additive class $[0]$ is realisable iff every defect section $\delta_\bullet(d): (H, +) \rightarrow (D, +)$ is a homomorphism. When some $\delta_\bullet(d)$ is not, $\text{im } \varphi^+$ is a torsor over $\ker o_\nu$ missing the origin.

Thus the residue enters through the defect δ , and the old “independent coordinate” story is replaced by a precise affine one: the realisable locus is a coset whose offset is the defect’s coboundary.

The method. Everything turns on one formal identity. In the trivial-kernel case the diagonal coboundary—the pair $(d_+\theta, d_\circ\theta_1)$ obtained from a single 1-cochain θ —satisfies the coupling *identically in θ* (Lemma 4.3). This is nothing but the inclusion $B_{\text{Sb}}^2 \subseteq Z_{\text{Sb}}^2$ restricted to the coupling component; but read as a statement about operators it shows o_ν kills B_+^2 and descends to H_+^2 , and the backward direction of $\text{im } \varphi^+ = \ker o_\nu$ then *constructs* the brace cocycle from a solution of the coupling rather than presupposing one—exactly the step the refuted argument got wrong. In the non-trivial-kernel case the same role is played by additivity of the two generating translations $\lambda_{\iota(d)}$ and $\lambda_{s(h)}$, which yield $(\star)$ and (F) respectively by a one-line expansion of a single skew-brace axiom instance.

Status of results. This is a capstone of a worked programme, and the parts have different strengths. We flag each result honestly at the point of use; in summary:

Result	Scope	Status
$[\nu] \perp [\beta]$ product	—	refuted (negative result, proved)
φ^+ descends	general	Lean-verified
$\Phi = (\varphi^+, \varphi^\circ)$ non-injective	witness	Lean-verified
o_ν a homomorphism, $\text{im } \varphi^+ = \ker o_\nu$	trivial kernel, general H	proved
$ G = 8$ collapse $o_\nu(b) = (1 + \nu)b \bmod 2D^{\widehat{\sigma}}$	trivial kernel	proved; Lean-verified
$\text{im } \varphi^+$ well defined on H_+^2	non-trivial kernel	proved (complex-free)
home $(D, \circ)$, twist $\widehat{\sigma}$; defect δ	non-trivial kernel	proved
kernel coupling $(\star)$; offset $= d_+[\delta_\bullet(d)]$	general H	proved
section coupling (F)	general H	proved
lock $(\lambda_1^D - 1)b \equiv 2(\nu\widehat{\sigma} - 1)$	$H = \mathbb{Z}/2$	proved (112/112); Lean-verified
$\text{im } \varphi^+ = \{[\beta] : (\star) \text{ solvable}\}$	$H = \mathbb{Z}/2$ / general H	proved / computed (34/34)
torsor dichotomy	general H	proved

The two open steps— $(\star) \Rightarrow (\text{F})$ -solvability for general H , and the homomorphism-law book-keeping for general H —are isolated in §7; neither touches the new content.

Credit. The reformulation “the coupling is affine; solvability is the vanishing of a class $o_\nu([\beta])$; $\text{im } \varphi^+ = \ker o_\nu$ ” is due to Rick [3], Remark 4 of the referee report that refuted the product claim. The present paper supplies the general- H operators, the descent via the linchpin identity, the non-trivial-kernel corrections (defect, home, twist), the split of the coupling into $(\star)$ and (F), the closed-form offset, and the torsor dichotomy. The general- H $(\star) \Rightarrow (\text{F})$ construction was developed jointly with Rick and is not claimed here.

Outline. §2 recalls skew-brace extensions, the good triplet, the Rathee–Yadav cochains and the two forgetful maps. §3 states and documents the refutation. §4 proves Theorem 1.1. §5 treats the non-trivial kernel: corrections, the complex-free well-definedness, the split coupling, the offset, the lock, and the torsor dichotomy (Theorem 1.2). §6 describes the Lean 4 certificates. §7 isolates the two open steps and closes with context.

2. PRELIMINARIES

2.1. Skew braces and the good triplet. Throughout, H is a skew brace (the *base*) with left action $\lambda^H : (H, \circ) \rightarrow \text{Aut}(H, +)$, $h \circ k = h +_H \lambda_h^H(k)$, and $(D, +)$ is an abelian group carrying a brace structure $\lambda^D : (D, \circ) \rightarrow \text{Aut}(D, +)$, $d \circ e = d + \lambda_d^D(e)$, with $(D, \circ)$ abelian. We call D a *trivial brace* when $\circ = +$ on D , i.e. $\lambda^D \equiv \text{id}$. Fix a skew-brace extension $0 \rightarrow D \rightarrow G \rightarrow H \rightarrow 0$ with D an ideal, and assume $(G, +)$ is abelian (so the additive conjugation μ below is trivial); this holds in all our examples and keeps μ out of the way.

Choose a normalised set-theoretic section $s : H \rightarrow G$, $s(0) = 0$. Following Rathee–Yadav [1, §2] it determines the *good triplet*

$$\nu_h = \lambda_{s(h)}|_D, \quad \mu_h(x) = -s(h) + x + s(h), \quad \widehat{\sigma}_h(x) = s(h)^{-1} \circ x \circ s(h), \quad (1)$$

where λ is G ’s brace action. Here $\nu : (H, \circ) \rightarrow \text{Aut}(D, +)$ is a homomorphism (the *residue*), $\mu : (H, +) \rightarrow \text{Aut}(D, +)$ is an anti-homomorphism (trivial, as $(G, +)$ is abelian), and $\widehat{\sigma} : (H, \circ) \rightarrow \text{Aut}(D, +)$ is the $\circ$ -conjugation. All three are independent of the choice of s [1, §2]. When D is trivial, $\circ = +$ on D and $\widehat{\sigma}$ coincides with the additive conjugation; in general $\widehat{\sigma}$ is the automorphism of $(D, \circ)$ by which H acts on the multiplicative kernel, and must be kept distinct from the additive data (§5).

Definition 2.1 (Intrinsic defect). $\delta_h := \mu_h^{-1} \nu_h \widehat{\sigma}_h - \text{id} \in \text{End}(D, +)$. With $\mu = \text{id}$, $\delta_h = \nu_h \widehat{\sigma}_h - \text{id}$.

The defect measures the failure of the collapse $\nu_h \widehat{\sigma}_h = \mu_h$ that holds automatically for a trivial kernel. It is additive, depends only on the triplet, and $\delta \equiv 0$ iff D is a trivial-brace ideal (Lemma 5.1).

2.2. The Rathee–Yadav cochains. Write $C^k = C^k(H, D)$ for the normalised k -cochains $H^k \rightarrow D$. To the section s , [1] attach two factor sets,

$$\beta(h_1, h_2) = -s(h_1 + h_2) + s(h_1) + s(h_2), \quad \bar{\tau}(h_1, h_2) = s(h_1 \circ h_2)^{-1} \circ s(h_1) \circ s(h_2), \quad (2)$$

the additive and multiplicative factor sets. A pair $(\beta, \bar{\tau})$ is a *skew-brace 2-cocycle*, $(\beta, \bar{\tau}) \in Z_{\text{Sb}}^2$, iff

- (a) $\beta \in Z_+^2 := Z_{\text{Gp}}^2((H, +), (D, +); \mu)$;
- (b) $\bar{\tau} \in Z_\circ^2 := Z_{\text{Gp}}^2((H, \circ), (D, \circ); \widehat{\sigma})$;
- (c) the coupling [1, (3.10)] binding $\bar{\tau}$ to β .

Modulo coboundaries B_{Sb}^2 this gives $H_{\text{Sb}}^2(H, D)$, which classifies the extensions inducing the triplet [1, §3]. We write $H_+^2 = Z_+^2 / B_+^2$.

2.3. The two forgetful maps. The additive factor set defines $\varphi^+ : H_{\text{Sb}}^2 \rightarrow H_+^2$, $[(\beta, \bar{\tau})] \mapsto [\beta]$, and symmetrically $\varphi^\circ : H_{\text{Sb}}^2 \rightarrow H_\circ^2$, $[(\beta, \bar{\tau})] \mapsto [\bar{\tau}]$. The following two facts frame the paper and carry machine certificates (§6).

Proposition 2.2 ([2, Thm. 3.7]; Lean-verified). *φ^+ is well defined: the coupling (c) ties the cocycles but never the coboundaries, so $[\beta]$ depends only on $[(\beta, \bar{\tau})]$.*

Proposition 2.3 ([2, W₄]; Lean-verified). *The product forgetful map $\Phi = (\varphi^+, \varphi^\circ)$ is not injective: there is a class with $\Phi([\Omega]) = (0, 0)$ yet $[\Omega] \neq 0$. Hence the extension is not determined by its additive and multiplicative types separately—the obstruction is irreducibly bilinear.*

Proposition 2.3 is what makes “compute $\text{im } \varphi^+$ ” a non-trivial question: knowing which additive classes occur is strictly less than knowing the extension, and the coupling is where the missing information lives. The rest of the paper computes $\text{im } \varphi^+$.

2.4. The finite criterion. For the explicit assemblies we use the Guarnieri–Vendramin criterion [4]: a set with an abelian group $(G, +)$ and a map $\lambda : G \rightarrow \text{Map}(G, G)$, $g \circ g' := g + \lambda_g(g')$, is a skew left brace iff (i) each $\lambda_g \in \text{Aut}(G, +)$ and (ii) $\lambda_{g \circ g'} = \lambda_g \lambda_{g'}$; then $(G, \circ)$ is automatically a group.

3. THE REFUTATION

We record the negative result, both because honesty demands it and because the correct obstruction is legible only against it.

3.1. The claim and the circular step. The refuted claim asserted that $[\nu]$ and $[\beta]$ are independent coordinates on H_{Sb}^2 : that for every residue, every additive class is realisable, so φ^+ is surjective and $\text{im } \varphi^+$ is residue-free. The argument fixed the triplet, wrote the coupling as an equation to be solved for the multiplicative datum, and claimed the system is homogeneous and residue-free after “reducing $\circ$ to $+$ and λ .”

The reduction is circular. “ $\circ$ is determined by $+$ and λ ” presupposes a brace action λ on the additive group built from β , for which $a \mapsto \lambda_a$ is a homomorphism for the resulting $\circ$. Constructing such a λ with a prescribed residue is the existence question; associativity of $\circ$ yields the cocycle conditions once a brace exists, but cannot manufacture the brace. That the *homogeneous* coupling is residue-free does not make the *inhomogeneous* solvability residue-free.

Proposition 3.1 (Refutation; proved). *The product decomposition $[\nu] \perp [\beta]$ is false. Concretely, $\text{im } \varphi^+$ depends on the residue already at $|G| = 8$, and the dependence is a genuine three-way coupling of $([\nu], \hat{\sigma}, [\beta])$ at $|G| = 16$.*

3.2. The two witnesses. We give the witnesses now as data; both re-appear in §§4–5 as corollaries of the repaired theorems.

Example 3.2 ($|G| = 8$). $H = \mathbb{Z}/2$, $D = \mathbb{Z}/4$ (trivial brace), $\mu = \text{id}$, $\hat{\sigma} = -\text{id}$. A normalised cochain is a single value $b = \beta(1, 1) \in D$. With $\hat{\sigma} = -1$ one has $D^{\hat{\sigma}} = \{0, 2\}$ and $2D^{\hat{\sigma}} = \{0\}$. The realisability of b is governed (Theorem 4.4) by $(1 + \nu)b \in 2D^{\hat{\sigma}}$:

$$\nu = +1 : 2b = 0 \iff b \in \{0, 2\} \iff [\beta] = 0; \quad \nu = -1 : 0 \in \{0\} \text{ always} \iff [\beta] \text{ free.}$$

So $\nu = +1$ forces $[\beta] = 0$: the additive group must be $\mathbb{Z}/2 \times \mathbb{Z}/4$, not $\mathbb{Z}/8$. The residue selects the realisable additive classes.

Example 3.3 ($|G| = 16$). $D = \mathbb{Z}/8$ with the non-trivial brace $\lambda_x^D = 5^x$ (so $L := \text{im } \lambda^D = \{1, 5\} \subsetneq \text{Aut}(D)$), $H = \mathbb{Z}/2$. Every realising extension satisfies the *lock*

$$(\lambda_d^D - \text{id})b = 2(\nu\hat{\sigma}(d) - d) \quad (d \in D), \quad \text{at } d = 1 : 4b \equiv 2(\nu\hat{\sigma} - 1) \pmod{8}, \quad (3)$$

so $[\beta] \neq 0$ (equivalently b odd) iff $\nu\hat{\sigma} \equiv 3 \pmod{4}$. The residue, the circle action, and the additive class are locked together through the product $\nu\hat{\sigma}$; no single two-way product reproduces the table.

Both witnesses were verified by two independent enumerations—Rick’s from-scratch regular-subgroup-of-holomorph computation and ours—and the refutation itself was reproduced from scratch [3]. What the earlier “corner” computation had actually shown is weaker than independence: that the decomposition class is not a *function* of $[\nu]$ (one fibre varies), which is consistent with—indeed implied by—coupling.

4. THE REPAIR, I: TRIVIAL KERNEL

Throughout this section D is a trivial brace and ν is a homomorphism.

4.1. The coupling operators. Read the Rathee–Yadav coupling (c) as an affine-linear equation in the multiplicative datum for fixed β . Define $\mathbb{Z}$ -linear operators $C^2 \rightarrow C^3$,

$$(P\tau)(h_1, h_2, h_3) = \mu_{\lambda_{h_1}(h_3)}(\tau(h_1, h_2)) - \tau(h_1, h_2 + h_3) + \tau(h_1, h_3), \quad (4)$$

$$(Q\beta)(h_1, h_2, h_3) = \nu_{h_1}(\beta(h_2, h_3)) + \mu_{h_1 \circ h_3}(\beta(h_1, -h_1)) - \beta(-h_1, h_1 \circ h_3) - \beta(h_1 \circ h_2, \lambda_{h_1}(h_3)), \quad (5)$$

where λ is the base brace’s action (fixed data). With $T : C^2 \rightarrow C^2$, $(T\bar{\tau})(h_1, h_2) = \nu_{h_1 \circ h_2}(\bar{\tau}(h_1, h_2))$ (so $\tau = T\bar{\tau}$), the coupling (c) reads

$$P(T\bar{\tau}) = Q\beta. \quad (6)$$

Both operators depend only on the triplet and on the group structure of H : *no brace is presupposed*.

Lemma 4.1 (Operators transcribe [1, (3.10)]; proved). *For every genuine skew brace G with trivial-brace ideal D and every section s , the honest cochains β, τ satisfy $P\tau = Q\beta$.*

Proof. Formula (5) is [1, (3.10)] with the sign of τ moved to the left; at $H = \mathbb{Z}/2$, μ trivial, one computes by hand $(P\tau) = 2t$ and $(Q\beta) = (1 + \nu)b$ with $t = \bar{\tau}(1, 1)$, $b = \beta(1, 1)$, recovering the coupling verbatim. The identity $P\tau = Q\beta$ was confirmed exhaustively on all genuine trivial-kernel braces for $H \in \{\mathbb{Z}/2, \mathbb{Z}/3, \mathbb{Z}/4\}$. $\square$

Let $A := P \circ T$ on $Z_\circ^2$; its image $A(Z_\circ^2) = P(T Z_\circ^2) \leq C^3$ is a subgroup, since P, T are linear and $Z_\circ^2$ is a group.

Definition 4.2. $o_\nu(\beta) := [Q\beta] \in C^3/A(Z_\circ^2)$ for $\beta \in Z_+^2$.

4.2. The linchpin identity. The entire descent rests on a single formal identity.

Lemma 4.3 (Linchpin; proved). *Let $\theta: H \rightarrow D$ with $\theta(0) = 0$ be arbitrary and set $\theta_1(h) := \nu_h^{-1}(\theta(h))$. Let $\beta := d_+\theta$ be the additive coboundary and $\bar{\tau} := d_\circ\theta_1$ the multiplicative coboundary, $(d_+\theta)(h_1, h_2) = \mu_{h_2}(\theta(h_1)) - \theta(h_1 + h_2) + \theta(h_2)$, $(d_\circ\theta_1)(h_1, h_2) = \hat{\sigma}_{h_2}(\theta_1(h_1)) - \theta_1(h_1 \circ h_2) + \theta_1(h_2)$. Then $P(T d_\circ\theta_1) = Q(d_+\theta)$ identically in θ .*

Proof. This is $B_{\text{Sb}}^2 \subseteq Z_{\text{Sb}}^2$ restricted to the coupling component: the diagonal coboundary $(d_+\theta, d_\circ\theta_1)$ lies in Z_{Sb}^2 , hence satisfies (6). Rathee–Yadav establish $B^2 \subseteq Z^2$ as a formal identity in θ and the triplet, requiring no brace to exist [1, §3]. We verified the identity exhaustively by machine over all θ and all good triplets harvested from genuine braces for $(H, D) = (\mathbb{Z}/2, \mathbb{Z}/4), (\mathbb{Z}/2, (\mathbb{Z}/2)^2), (\mathbb{Z}/3, \mathbb{Z}/3), (\mathbb{Z}/4, \mathbb{Z}/4)$. $\square$

4.3. The main theorem.

Theorem 4.4 (Main, trivial kernel; proved). *Fix the good triplet with D a trivial brace and ν a homomorphism.*

- (1) $o_\nu(d_+\theta) = 0$ for every θ ; hence o_ν descends to a map $o_\nu: H_+^2 \rightarrow C^3/A(Z_\circ^2)$, $[\beta] \mapsto [Q\beta]$.
- (2) o_ν is a group homomorphism.
- (3) $\text{im } \varphi^+ = \ker o_\nu$: an additive class $[\beta]$ extends to a full skew-brace cocycle inducing the triplet iff $o_\nu([\beta]) = 0$.

Proof. (1) By Lemma 4.3, $Q(d_+\theta) = P(T d_\circ\theta_1) \in A(Z_\circ^2)$, since $d_\circ\theta_1 \in Z_\circ^2$ (a coboundary is a cocycle). Thus $[Q(d_+\theta)] = 0$, so o_ν kills B_+^2 and factors through $H_+^2 = Z_+^2/B_+^2$. The target is fixed by the triplet, so o_ν is well defined on classes.

(2) Each summand of (5) is additive in β , because $\nu_{h_1}, \mu_{h_1 \circ h_3} \in \text{Aut}(D, +)$ and reindexing is additive; and $A(Z_\circ^2)$ is a subgroup. Hence $[\beta] \mapsto [Q\beta]$ is a homomorphism.

(3) ($\subseteq$) If $[\beta] = \varphi^+[(\beta, \bar{\tau})]$ then $(\beta, \bar{\tau}) \in Z_{\text{Sb}}^2$, so by (c)/(6) $Q\beta = P(T\bar{\tau})$ with $\bar{\tau} \in Z_\circ^2$ (from (b)); hence $Q\beta \in A(Z_\circ^2)$, i.e. $o_\nu([\beta]) = 0$.

($\supseteq$) If $o_\nu([\beta]) = 0$ then $Q\beta \in A(Z_\circ^2)$: there is $\bar{\tau} \in Z_\circ^2$ with $P(T\bar{\tau}) = Q\beta$. The pair $(\beta, \bar{\tau})$ then satisfies (a), (b), and (c)/(6); by the Rathee–Yadav characterisation $(\beta, \bar{\tau}) \in Z_{\text{Sb}}^2$, whence $[\beta] = \varphi^+[(\beta, \bar{\tau})] \in \text{im } \varphi^+$. This direction *constructs* the cocycle rather than presupposing one—the circular step of §3 is thereby avoided. $\square$

Proposition 4.5 (Cross-validation; computed). *For $H = \mathbb{Z}/2$ with $D = \mathbb{Z}/4$ (all 4 triplets) and $D = (\mathbb{Z}/2)^2$ (all 16 triplets), and for $H = \mathbb{Z}/3$ with $D = \mathbb{Z}/3$, the abstractly computed $\ker o_\nu \subseteq H_+^2$ equals the $\text{Aut}(D)$ -saturated set of additive classes realised by genuine skew braces with the given triplet, from an independent regular-subgroup-of-holomorph enumeration. For $D = (\mathbb{Z}/2)^2$ the obstruction is genuinely nonzero for many triplets, cutting $|H_+^2| = 4$ down to $|\ker o_\nu| = 2$.*

Example 3.2 is now a corollary: with $\hat{\sigma} = -1$, T acts trivially, $A(Z_\circ^2) = 2D^{\hat{\sigma}}$, and (5) collapses to $o_\nu(b) = 0 \iff (1 + \nu)b \in 2D^{\hat{\sigma}}$.

5. THE REPAIR, II: NON-TRIVIAL KERNEL

We now drop the triviality of D . Two features of the trivial-kernel treatment were using that hypothesis silently; each must be corrected before the obstruction survives.

5.1. Two corrections.

Lemma 5.1 (Intrinsic defect; proved). *Rathee–Yadav’s identity (3.1) holds with no triviality hypothesis: for $x \in D$, $\lambda_x(s(h)) - s(h) = -x + \mu_h^{-1}(\nu_h \hat{\sigma}_h(x))$. Consequently $\delta_h(x) = \lambda_x(s(h)) - s(h)$, $\delta_h \in \text{End}(D, +)$ is determined by the triplet alone, and $\delta \equiv 0$ iff D is a trivial-brace kernel.*

Proof. The displayed identity rearranges to $\mu_h(x + \delta_h(x)) = \nu_h \hat{\sigma}_h(x)$, i.e. $\delta_h = \mu_h^{-1} \nu_h \hat{\sigma}_h - \text{id}$ (Definition 2.1), a difference of the automorphism $\mu_h^{-1} \nu_h \hat{\sigma}_h$ and id in the abelian ring $\text{End}(D, +)$, hence additive. It is manifestly a function of $(\nu, \mu, \hat{\sigma})$. Vanishing is equivalent to $\nu_h \hat{\sigma}_h = \mu_h$, i.e. $\lambda_x(s(h)) = s(h)$ for all x, h , the trivial-kernel collapse. $\square$

Triviality forces the left side of (3.1) to vanish; this is exactly the collapse Rathee–Yadav use to simplify their action. The defect is the measurable amount by which it fails.

Theorem 5.2 (Home and twist; proved). *The multiplicative factor set $\bar{\tau}$ of (2) is the factor set of the group extension $0 \rightarrow (D, \circ) \rightarrow (G, \circ) \rightarrow (H, \circ) \rightarrow 0$; its kernel is $(D, \circ)$ and the conjugation action of H on it is $\hat{\sigma}$. Hence $\bar{\tau}$ is a 2-cocycle of $(H, \circ)$ valued in $(D, \circ)$, twisted by $\hat{\sigma}$ —not a $(D, +)$ -cocycle, and not twisted by the additive conjugation. Over $(D, +)$ the multiplicative differential fails $d^2 = 0$.*

Proof. $\bar{\tau}$ is the factor set of $(G, \circ)$ over $(H, \circ)$ with abelian kernel $(D, \circ)$, so it is a $\circ$ -cocycle for the conjugation action $\hat{\sigma}$, by the standard classification of group extensions. *Necessity of $(D, \circ)$:* the naive $(D, +)$ multiplicative differential has $d^2 \neq 0$ on 2/16 good triplets with $D = \mathbb{Z}/8$. *Necessity of $\hat{\sigma}$ over the additive conjugation σ_+ :* the coupling constant $2(\nu \hat{\sigma} - 1)$ of §5.4 holds 104/104 with $\hat{\sigma}$ and only 80/104 with σ_+ , the 24 failures being exactly the $D = \mathbb{Z}/2 \times \mathbb{Z}/8$ braces with $\nu \not\equiv 1 \pmod{4}$. For a trivial kernel $\circ = +$ and $\hat{\sigma} = \sigma_+$, recovering Rathee–Yadav. $\square$

5.2. $\text{im } \varphi^+$ is well defined, complex-free. This is the load-bearing structural fact, and it needs no non-trivial-kernel cochain complex. Say $[\beta] \in H_+^2$ is *realisable* for the fixed triplet if some representative β is the additive factor set of a genuine extension inducing the triplet; $\text{im } \varphi^+$ is the set of realisable classes.

Theorem 5.3 (Well-definedness; proved). *$\text{im } \varphi^+$ is a union of B_+^2 -cosets; realisability of β depends only on $[\beta] \in H_+^2$.*

Proof. Let β be realised by (G, s) with multiplicative factor set $\bar{\tau}$. Fix a 1-cochain $\theta: H \rightarrow D$, $\theta(0) = 0$, and define a new normalised section of the *same* G by $s'(h) := \theta(h) + s(h)$. Using $(G, +)$ abelian, its additive factor set is

$$\beta'(h_1, h_2) = -s'(h_1 + h_2) + s'(h_1) + s'(h_2) = \beta(h_1, h_2) + [\theta(h_1) + \theta(h_2) - \theta(h_1 + h_2)] = \beta + d_+ \theta,$$

and its multiplicative factor set $\bar{\tau}'$ differs from $\bar{\tau}$ by the $(D, \circ)$ -coboundary $d_\circ \theta_1$ ($\theta_1(h) = \nu_h^{-1}(\theta(h))$). Both $(\beta, \bar{\tau})$ and $(\beta', \bar{\tau}')$ are honest factor sets of the *same* extension, so both realise their additive classes; in particular $\beta' = \beta + d_+ \theta$ is realisable. Since the triplet is independent of the section [1, §2], β' realises its class within the same triplet-indexed H_+^2 . As θ ranges over all 1-cochains, $d_+ \theta$ ranges over B_+^2 , so every representative of $[\beta]$ is realisable. $\square$

This replaces, in the non-trivial regime, the trivial-kernel “ o_ν descends because $B^2 \subseteq Z^2$ ” step, and it is valid verbatim without assuming a well-behaved non-trivial-kernel cohomology.

5.3. The assembly and the split coupling. Coordinatise $G \cong D \times H$ by $x \mapsto (x - s(qx), qx)$, q the quotient map. A genuine skew brace inducing the triplet becomes, on $D \times H$,

$$(a, h_1) + (b, h_2) = (a + b + \beta(h_1, h_2), h_1 +_H h_2), \quad (7)$$

$$(a, h_1) \circ (b, h_2) = (a, h_1) + \lambda_{(a, h_1)}(b, h_2), \quad (8)$$

with the left translation assembled from

$$\lambda_{s(h)}(b, k) = (\nu_h b + f(h, k), \lambda_h^H k), \quad f(h, k) = \beta(-h, h \circ k) - \beta(h, -h) + \nu_{h \circ k} \bar{\tau}(h, k), \quad (9)$$

$$\lambda_{\iota(d)}(b, k) = (\lambda_d^D b + \delta_k(d), k), \quad (d \in D), \quad (10)$$

$$\lambda_{(a, h)} = \lambda_{\iota(a')} \circ \lambda_{s(h)}, \quad a' = (\nu_h \widehat{\sigma}_h)^{-1}(a) = (\text{id} + \delta_h)^{-1}(a). \quad (11)$$

Conversely, any pair $(\beta, \bar{\tau})$ of normalised 2-cochains *assembles* to a structure (7)–(11); the question is when the assembly is a skew brace. The assembly reproduces every genuine non-trivial-kernel brace exactly (33/33 across $D \in \{\mathbb{Z}/4, \mathbb{Z}/8\}$, $H = \mathbb{Z}/2$, and $D = \mathbb{Z}/4$, $H = \mathbb{Z}/3$). By the finite criterion the assembly is a brace iff every λ_g is additive and $\lambda_{g \circ g'} = \lambda_g \lambda_{g'}$; we analyse the generating translations $\lambda_{\iota(d)}$ and $\lambda_{s(h)}$ in turn.

Lemma 5.4 (Kernel coupling $(\star)$; proved, general H). *If the assembly is a skew brace then each $\lambda_{\iota(d)}$ is an additive automorphism, and this holds iff*

$$(\lambda_d^D - \text{id}) \beta(k, k') = \delta_k(d) + \delta_{k'}(d) - \delta_{k+H k'}(d) \quad (d \in D, k, k' \in H). \quad (\star)$$

Proof. Expand $\lambda_{\iota(d)}((b, k) + (b', k')) = \lambda_{\iota(d)}(b, k) + \lambda_{\iota(d)}(b', k')$ by (7), (10). The H -components agree; since $\lambda_d^D \in \text{Aut}(D, +)$ the terms $\lambda_d^D(b+b')$ cancel, and the D -components give exactly $(\star)$. Conversely $(\star)$, with $\lambda_d^D \in \text{Aut}(D, +)$ and $\delta_k \in \text{End}(D, +)$, makes $\lambda_{\iota(d)}$ additive; it is bijective since λ_d^D is and the H -coordinate is fixed. The derivation uses no triviality and no property of $\bar{\tau}$. $\square$

Remark 5.5 (Recovery of the lock). At $H = \mathbb{Z}/2$, $k = k' = 1$ (so $k +_H k' = 0$, $\delta_0 = 0$), $(\star)$ reads $(\lambda_d^D - \text{id})b = 2\delta_1(d)$ with $b = \beta(1, 1)$ —the lock (3) of Example 3.3, here the $|H| = 2$ slice of a single general- H identity.

Corollary 5.6 (Closed-form offset; proved, general H). *Write $\psi^d \in C^1(H, D)$ for $\psi_k^d := \delta_k(d)$, linear in d . Then $(\star)$ is the affine equation*

$$(\lambda_d^D - \text{id}) \beta = d_+ \psi^d \quad (\forall d \in D), \quad (12)$$

whose left side is $\mathbb{Z}$ -linear in β and whose right side is the β -independent additive coboundary $c^d := d_+[\delta_\bullet(d)]$. Thus the offset carried by o_ν is, in closed form and for arbitrary H , the coboundary of the defect section.

Proof. With $\mu = \text{id}$, $(d_+ \psi^d)(k, k') = \psi_k^d - \psi_{k+H k'}^d + \psi_{k'}^d = \delta_k(d) - \delta_{k+H k'}(d) + \delta_{k'}(d)$, the right side of $(\star)$; the left side is linear since $\lambda_d^D - \text{id} \in \text{End}(D, +)$ acts pointwise. c^d depends only on the triplet. $\square$

Definition 5.7 (The obstruction, non-trivial kernel). In the non-trivial-kernel regime we write $o_\nu([\beta])$ for the obstruction to solving $(\star)$ at the cochain level for a representative of $[\beta]$; by Theorem 5.3 this is well defined on H_+^2 , and $o_\nu([\beta]) = 0$ is precisely the condition of Theorem 5.13. By Corollary 5.6 it is *affine*, with linear part $[\beta] \mapsto (\lambda_\bullet^D - \text{id})[\beta]$ and constant offset $c^\bullet = d_+[\delta_\bullet(\cdot)]$ —in contrast to the trivial-kernel case ($\delta \equiv 0$), where o_ν is the linear map of Theorem 4.4.

Corollary 5.8 (Cohomological necessity). *If $[\beta] \in \text{im } \varphi^+$ then $(\lambda_d^D - \text{id})_*[\beta] = 0$ in H_+^2 for every d . This is necessary but strictly weaker than $(\star)$: the offset c^d must be matched at the cochain level by a single representative, not merely up to coboundary. (At $D = \mathbb{Z}/8$, $H = \mathbb{Z}/2$, the cohomological condition admits both classes while $\text{im } \varphi^+$ is a single class—the torsor phenomenon of §5.6.)*

Proof. $(\lambda_d^D - \text{id})\beta = d_+ \psi^d$ is a coboundary, so its class vanishes; strictness is the cited computation. $\square$

Lemma 5.9 (Section coupling (F); proved, general H). *Additivity of $\lambda_{s(h)}$ holds iff*

$$f(h, k) + f(h, k') - f(h, k +_H k') = \nu_h \beta(k, k') - \beta(\lambda_h^H k, \lambda_h^H k') \quad (h, k, k' \in H), \quad (\text{F})$$

with f as in (9). For a trivial kernel $(\star)$ is vacuous and (F) is the Rathee–Yadav coupling (3.10); (F) is its non-trivial-kernel form.

Proof. Expand $\lambda_{s(h)}((b, k) + (b', k')) = \lambda_{s(h)}(b, k) + \lambda_{s(h)}(b', k')$ by (7), (9). The H -components agree as $\lambda_h^H \in \text{Aut}(H, +)$; the D -components give (F). In the trivial-kernel case (F) coincides with (3.10) after substituting $f = \beta(-h, h \circ k) - \beta(h, -h) + \tau(h, k)$ and using $\beta \in Z_+^2$; this matches $P\tau = Q\beta$ on all trivial-kernel braces for $H \in \{\mathbb{Z}/2, \mathbb{Z}/3, \mathbb{Z}/4\}$. $\square$

Lemma 5.10 (Kernel homomorphism, cyclic D ; proved). *If $(D, +)$ is cyclic then $d \mapsto \lambda_{\iota(d)}$ is a homomorphism $(D, \circ) \rightarrow \text{Aut}(G, +)$.*

Proof. Both sides fix the H -coordinate. On D -coordinates equality reduces to $\lambda_d^D \delta_k(e) = \delta_k(\lambda_d^D e)$, i.e. $[\lambda_d^D, \delta_k] = 0$ in $\text{End}(D, +)$, using $\lambda_{d \circ e}^D = \lambda_d^D \lambda_e^D$ and $\delta_k(d \circ e) = \delta_k(d) + \delta_k(\lambda_d^D e)$. For cyclic D , $\text{End}(D, +) = \mathbb{Z}/|D|$ is commutative. (For non-cyclic D this commutation is an extra hypothesis; all non-trivial-kernel families in evidence have cyclic D .) $\square$

Theorem 5.11 (Decomposition of the axioms). *Assume D cyclic. The assembly of $(\beta, \bar{\tau})$ is a skew brace iff*

$$(\text{add}) \beta \in Z_+^2, \quad (\text{mult}) \bar{\tau} \in Z_o^2, \quad (\text{F}) (\text{F}), \quad (\star) (\star).$$

Once each generating translation is additive (i.e. $(\star)$ and (F) hold) and $\bar{\tau} \in Z_o^2$, the homomorphism law $\lambda_{g \circ g'} = \lambda_g \lambda_{g'}$ is automatic. Only $(\star)$ involves β without $\bar{\tau}$; only (F) couples $\bar{\tau}$ to β .

Proof and status. (add) is the group law for (7); (mult) is the group law for the multiplicative factor set (kernel $(D, \circ)$, action $\hat{\sigma}$; Theorem 5.2). Necessity of $(\star)$ and (F) is Lemmas 5.4, 5.9. For sufficiency one promotes generator-additivity to the homomorphism law on $(G, \circ)$, which is generated by $\iota(D) \cup s(H)$ subject to $\iota(d) \circ \iota(e) = \iota(d \circ e)$ (Lemma 5.10), $s(h) \circ s(k) = \iota(\bar{\tau}(h, k)) \circ s(h \circ k)$ (using (mult)), and the $\hat{\sigma}$ -conjugation relation; λ respects each relation exactly when (F), $(\star)$, (mult) hold. This reduction is proved at $H = \mathbb{Z}/2$ and machine-verified exhaustively there (brace $\Leftrightarrow$ (F) $\wedge$ $(\star)$ with zero exceptions, $D \in \{\mathbb{Z}/4, \mathbb{Z}/8\}$); the general- H relation bookkeeping is the mechanical residue of §7. $\square$

5.4. The lock and the explicit $H = \mathbb{Z}/2$ case. Write $u := s(1)$, $b := \beta(1, 1) = 2u$, $t := \bar{\tau}(1, 1) = u \circ u$.

Theorem 5.12 (The lock, by hand; proved, 112/112). *For $H = \mathbb{Z}/2$ and every $x \in D$,*

$$(\lambda_x^D - \text{id})b = 2\delta_1(x) = 2(\nu_1 \hat{\sigma}_1(x) - x) \quad \text{in } (D, +); \quad (13)$$

at a generator, $(\lambda_1^D - 1)b \equiv 2(\nu \hat{\sigma} - 1) \pmod{|D|}$.

Proof. Apply the skew-brace axiom $a \circ (b' + c') = (a \circ b') - a + (a \circ c')$ with $a = x \in D$ and $b' = c' = u$. As $u + u = b \in D$, the left side is $x \circ b = x + \lambda_x^D(b)$. For the right side, $x \circ u = x + \lambda_x(u) = x + (u + \delta_1(x))$ by Lemma 5.1; adding two copies with $(G, +)$ abelian gives $2(x \circ u) - x = x + 2\delta_1(x) + b$. Equating and cancelling $x + b$ yields $\lambda_x^D(b) - b = 2\delta_1(x)$, which is (13). The derivation is general; it is verified on all 112 genuine non-trivial-kernel braces with $H = \mathbb{Z}/2$ ($D \in \{\mathbb{Z}/4, \mathbb{Z}/8\}$). $\square$

For $D = \mathbb{Z}/8$ ($\lambda_1^D = 5$) this is $4b \equiv 2(\nu \hat{\sigma} - 1) \pmod{8}$, so $[\beta] \neq 0$ (i.e. b odd) iff $\nu \hat{\sigma} \equiv 3 \pmod{4}$ —Example 3.3.

5.5. The main biconditional.

Theorem 5.13 (Non-trivial kernel). *Assume D cyclic and ν a homomorphism. An additive class $[\beta] \in H_+^2$ is realisable iff some representative β admits $\bar{\tau} \in Z_\circ^2$ satisfying $(\star)$ and (F); and then*

$$\text{im } \varphi^+ = \{[\beta] \in H_+^2 : (\star) \text{ is solvable at the cochain level for some representative}\}.$$

The offset carried by o_ν is $c^d = d_+[\delta_\bullet(d)]$ (Corollary 5.6). This is proved for $H = \mathbb{Z}/2$ (finite construction) and verified computationally for all 34 homomorphism-triplets with $H = \mathbb{Z}/2$ ($D \in \{\mathbb{Z}/4, \mathbb{Z}/8\}$) and $H = \mathbb{Z}/3$ ($D = \mathbb{Z}/9$), i.e. 86 genuine braces: $(\star)$ holds on 86/86 and predicts $\text{im } \varphi^+$ on 34/34 triplets.

Proof. Forward ($\subseteq$): a realising brace gives β with $(\star)$ (Lemma 5.4) and $\bar{\tau} \in Z_\circ^2$ with (F) (Lemmas 5.9, 5.11). Reverse ($\supseteq$): given β with $(\star)$, choose $\bar{\tau} \in Z_\circ^2$ solving (F); Theorem 5.11 then yields a brace, so $[\beta]$ is realised, constructively. The reverse rests on (F) being solvable once $(\star)$ holds; this is finite at $H = \mathbb{Z}/2$ and verified on all 34 triplets, and is the single step left open for general H (§7). $\square$

5.6. The torsor dichotomy.

Theorem 5.14 (Torsor dichotomy; proved, general H). *The split additive class $[0]$ is realisable iff, for every $d \in D$, the defect section $\delta_\bullet(d): (H, +) \rightarrow (D, +)$, $k \mapsto \delta_k(d)$, is a group homomorphism (equivalently $c^d = d_+[\delta_\bullet(d)] = 0$). When some $\delta_\bullet(d)$ is not a homomorphism, $0 \notin \text{im } \varphi^+$ and $\text{im } \varphi^+$ is a torsor over $\ker o_\nu$ without basepoint. At $H = \mathbb{Z}/2$ this reads $2(\nu\hat{\sigma} - 1) = 0$: the witness $D = \mathbb{Z}/8$ ($L = \{1, 5\} \subsetneq \text{Aut } D$) is a torsor at $\nu\hat{\sigma} \equiv 3 \pmod{4}$, while the counter $D = \mathbb{Z}/4$ ($L = \text{Aut } D$) forces $c \equiv 0$ and is always based.*

Proof. Put $\beta = 0$ in $(\star)$: $0 = d_+\psi^d$ for all d , i.e. $\psi^d = \delta_\bullet(d)$ is a 1-cocycle, a homomorphism (with $\mu = \text{id}$). Realisability of $[0]$ requires $(\star)$ for a representative of $[0]$; changing representative alters the left side of $(\star)$ by $(\lambda_d^D - \text{id})d_+\theta$ while the offset c^d is fixed, so $[0]$ is realisable iff $c^d = 0$. At $H = \mathbb{Z}/2$, $\delta_\bullet(d)$ has the single value $\delta_1(d)$ and $c^d = 2\delta_1(d) = 2(\nu\hat{\sigma} - 1)d$. $\square$

Thus the no-basepoint phenomenon of Example 3.3 is intrinsic: it is the failure of the defect to be a homomorphism, which happens exactly when $L := \text{im } \lambda^D \subsetneq \text{Aut}(D)$ and $\nu\hat{\sigma}$ is non-sole (in the witness, $L = \{1, 5\}$). The residue is woven into realisability through δ ; there is no independent “ $[\nu] \perp [\beta]$ ” coordinate, only this affine coupling.

6. THE LEAN LAYER

Three finite, load-bearing facts carry machine certificates in Lean 4 core (Mathlib-free, v4.30.0), each built honestly from the operations—no hardcoded tables—and closed by **decide** or by explicit computation. We state what is certified and, equally importantly, its scope.

- φ^+ **descends** (Proposition 2.2). The additive component of a skew-brace coboundary is itself an additive coboundary—a pure associativity regrouping in $(G, +)$, using no $\tau, \nu, \hat{\sigma}$ —so φ^+ is well defined. Axiom footprint **propext** only.
- Φ **non-injective** (Proposition 2.3). On $G = \mathbb{Z}/2 \times \mathbb{Z}/4$ with the witness brace $a \circ b = (a_1 + b_1, a_2 + b_2 + 2a_1b_1)$, the cochain values $\beta(1, 1) = (0, 2y)$ and $\tau(1, 1) = (0, 2y + 2)$ are computed honestly from the operations over the four sections $y \in \mathbb{Z}/4$. Then $[\beta] = 0 \iff y \in \{0, 2\}$ and $[\tau] = 0 \iff y \in \{1, 3\}$; the two loci are disjoint and nonempty, so no section kills both, giving a class with $\Phi([\Omega]) = (0, 0)$ yet $[\Omega] \neq 0$. The obstruction is irreducibly bilinear.

- **The $|G| = 16$ lock and torsor dichotomy** (Theorems 5.12, 5.14 at $H = \mathbb{Z}/2$). On $D = \mathbb{Z}/8$ with $\lambda_1^D = 5$ (so $L = \{1, 5\} \subsetneq \text{Aut } D$) the lock $4b \equiv 2(\nu\hat{\sigma} - 1) \pmod{8}$, its descent to $H_+^2 = D/2D$, and the offset dichotomy (offset $\neq 0 \iff \nu\hat{\sigma} \equiv 3 \pmod{4}$) = torsor without basepoint) are decided outright; the lock holds as a necessary condition on all 12 section-invariant representatives (24 braces) of the holomorph enumeration. The counter $D = \mathbb{Z}/4$ ($L = \text{Aut } D$) has offset $\equiv 0$ always, forcing $\text{im } \varphi^+ = \{[0]\}$, based. Depends on *no* axioms (pure **decide**).

Each certificate covers exactly its finite witness: they do not re-encode the Rathee–Yadav characterisation, the abstract operators, or the general- H offset. Their role is to pin the arithmetic crux of the obstruction—that the loci are genuinely disjoint, that the lock is genuinely satisfied—beyond the reach of a transcription error.

7. OPEN STEPS AND OUTLOOK

7.1. Two isolated open steps. Theorem 5.13 is unconditional at $H = \mathbb{Z}/2$ and computationally confirmed on all 34 triplets tested; two mechanical residues remain for general H , both isolated and machine-covered on the tested range.

- (O1) $(\star) \Rightarrow (\text{F})$ -*solvability, general H* . The reverse of Theorem 5.13 needs: if β satisfies $(\star)$ then (F) is solvable in $\bar{\tau} \in Z_o^2$. This is one inhomogeneous $\mathbb{Z}$ -linear condition; it is finite at $H = \mathbb{Z}/2$ and holds on all 34 homomorphism-triplets tested. A general proof requires identifying the image of $\bar{\tau} \mapsto (\text{F-residual})$ and checking $(\star)$ lands in it. This step is being developed jointly with Rick.
- (O2) *Homomorphism-law bookkeeping, general H* . The sufficiency half of Theorem 5.11 (relation-checks for λ on the generators of $(G, \circ)$) is finite and machine-exhaustive at $H = \mathbb{Z}/2$; the general- H term-by-term reduction is routine but not written out.

Neither step touches the new content: the closed-form offset $c^d = d_+[\delta_\bullet(d)]$ (Corollary 5.6), the general- H kernel coupling $(\star)$ (Lemma 5.4), and the torsor dichotomy (Theorem 5.14) are unconditional. A genuine $H = \mathbb{Z}/3$, D -non-trivial example exercising the general- H prediction needs a larger D (e.g. $\mathbb{Z}/9$ or $(\mathbb{Z}/3)^2$ carrying a non-trivial brace with $\mathbb{Z}/3 \hookrightarrow \text{Aut}(D, +)$); the smallest candidate on $\mathbb{Z}/4$ yields a ν that is not a homomorphism, outside the hypotheses.

7.2. What the repair buys. The obstruction o_ν answers the forgetful-image question completely for a trivial kernel and up to two isolated steps for a non-trivial one, and it does so with a single, cohomologically meaningful object: a class whose vanishing is solvability, whose linear part is the kernel action $\lambda_d^D - \text{id}$ on H_+^2 , and whose offset is the coboundary of the intrinsic defect. The affine structure explains the empirical “torsor without basepoint” directly: the split extension exists iff the defect is a homomorphism. We regard the arc of this paper—a plausible product decomposition, refuted one review-hop away, and rebuilt as the correct affine obstruction—as the intended shape of the result, not an accident of its history. The false co-ordinate and the true coupling are two readings of the same data; only the second survives computation.

The obstruction here is cohomological, living on the H^2 side of skew-brace (co)homology; it is complementary to the homology development of Gran, Letourmy and Vendramin [5], whose Hopf formulae compute the low-degree homology of skew braces, and we see no overlap between the two.

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REFERENCES

- [1] M. Rathee and M. K. Yadav, *Skew brace extensions, second cohomology and complements*, arXiv:2601.12371.
- [2] MacBeth, *The additive forgetful map on skew-brace H^2 descends, and the product forgetful map is not injective* (internal note, Kodamai; the W_4 separation), 2026; machine-checked in Lean 4 (`additive-forgetful-descends, w4-noninjective`).
- [3] Rick, *Referee report: the $[\nu] \perp [\Omega]$ corner—cross-equation (c) is not auto-solvable; residue-dependent obstructions at $|G| = 8$ and $|G| = 16$* , 2026; Remark 4 (the affine reformulation in $\varphi^+ = \ker o_\nu$).
- [4] L. Guarnieri and L. Vendramin, *Skew braces and the Yang–Baxter equation*, Math. Comp. **86** (2017), 2519–2534.
- [5] M. Gran, T. Letourmy and L. Vendramin, *Hopf formulae for homology of skew braces*, arXiv:2409.18056.

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Email address: `neil@kodamai.com`